

EX.NO: 7

IMPLEMENTATION OF NAÏVE BAYES CLASSIFICATION ALGORITHM

AIM:

To classify the new instance (new sample) to a target data using Naïve Bayes classification algorithm in R Studio.

PROCEDURE:

1. Install the necessary package in R Studio such as “e1071”
2. Set the current working directory
3. Save the dataset in the current working directory
4. Read the dataset in the form of csv file using **read.csv()**.
5. Finally predict the category for the new instance using Naïve Bayes classifier.

SOURCE CODE:

```
# Loading package
library(e1071)
library(caTools)
library(caret)

# Splitting data
into train # and
test data

split <- sample.split(iris, SplitRatio
```

```
= 0.7) train_cl <- subset(iris, split  
== "TRUE") test_cl <- subset(iris,  
split == "FALSE") # Feature
```

Scaling

```
train_scale <- scale(train_cl[,  
1:4]) test_scale <-  
scale(test_cl[, 1:4])
```

Fitting Naive Bayes Model

to training dataset

set.seed(120)

Setting Seed

classifier_cl <- naiveBayes(Species ~ ., data = train_cl)

classifier_cl # Predicting on test data'

y_pred <- predict(classifier_cl, newdata = test_cl)

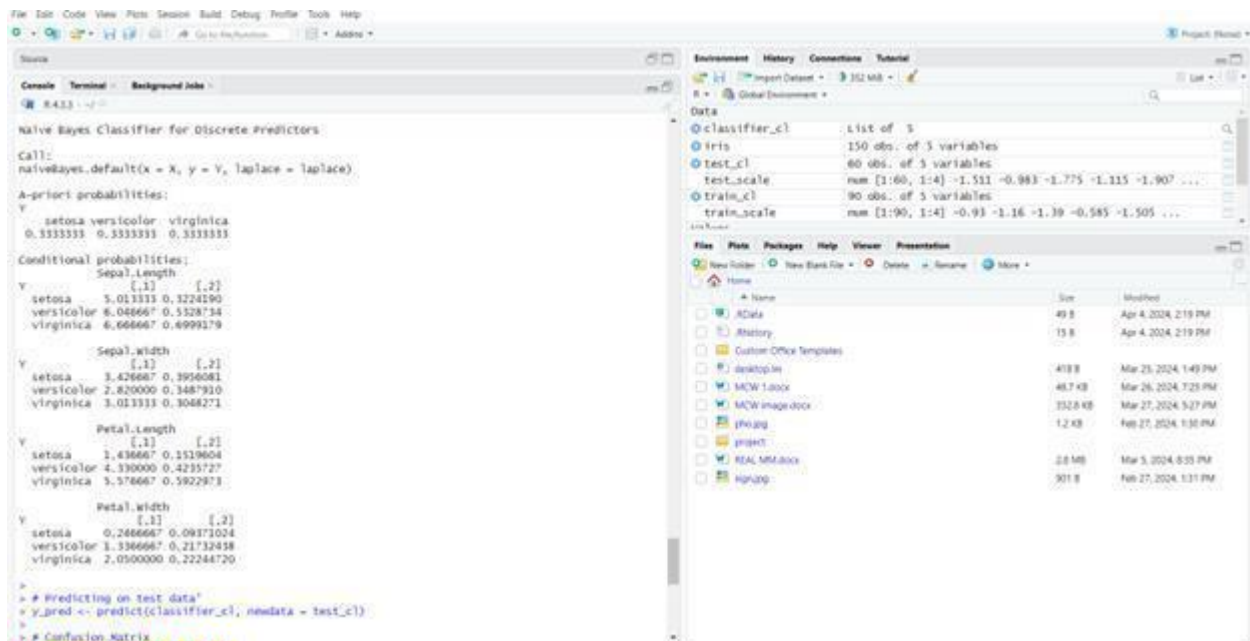
Confusion Matrix

cm <- table(test_cl\$Species,

y_pred) cm # Model Evaluation

confusionMatrix(cm)

OUTPUT:



```
Naive Bayes Classifier for Discrete Predictors
Call:
naiveBayes.default(x = X, y = Y, laplace = laplace)

A-priori probabilities:
Y
  setosa versicolor virginica 
0.3333333 0.3333333 0.3333333 

Conditional probabilities:
Sepal.Length
Y      [,1]      [,2]
setosa  5.013333 0.3224190
versicolor 8.046667 0.5328734
virginica 6.666667 0.6999179

Petal.Length
Y      [,1]      [,2]
setosa  3.426667 0.3910081
versicolor 2.820000 0.3487910
virginica 3.013333 0.3046271

Petal.Width
Y      [,1]      [,2]
setosa  1.436667 0.1519604
versicolor 4.330000 0.4235727
virginica 5.576667 0.5922973

# Predicting on test data
# y_pred <- predict(classifier_cl, newdata = test_cl)
# Confusion_Matrix
```

RESULT:

Thus, the new instance has been classified to a target attribute using Naïve Bayes classification algorithm.